

SMART WORKER HEALTH AND SAFETY MONITORING SYSTEM FOR CONSTRUCTION INDUSTRIES

ASSIGNMENT - 11

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PROJECT OVERVIEW

The Worker Health and Safety Monitoring System is a smart safety system designed to continuously monitor the working conditions of construction workers and identify potentially unsafe situations. Construction sites involve several hazards, including high temperature, excessive humidity, falls, and other emergency conditions that can affect worker safety and productivity.

The system uses sensors to collect safety-related parameters in real time and sends the data to a microcontroller for processing. The measured values are compared with predefined safe limits to determine whether the worker is operating under normal or unsafe conditions. When an abnormal condition is detected, the system automatically generates an alert through a buzzer, LED indicator, or display.

The system reduces the need for continuous manual supervision and helps workers and supervisors respond quickly to dangerous situations. The monitoring data can also be stored or transmitted through IoT technology for remote observation and future analysis. Overall, the proposed system provides a low-cost and automated approach to improving safety awareness and reducing risks in construction industries.

PROBLEM STATEMENT

Construction workers are exposed to various workplace hazards that can affect their health and safety during daily operations. High temperature, excessive humidity, unsafe environmental conditions, and sudden emergency situations can increase the risk of accidents and health problems. Traditional safety monitoring methods often depend on manual supervision, which may not provide continuous real-time information about changing conditions.

Delayed detection of unsafe conditions can make it difficult for workers and supervisors to take immediate preventive action. Therefore, there is a need for an automated monitoring system that continuously observes important safety parameters and generates alerts whenever the conditions exceed predefined safe limits. The proposed system aims to improve worker safety by providing timely warnings and reducing dependence on continuous manual monitoring.

Therefore, an automated Worker Health and Safety Monitoring System is required to continuously monitor important safety parameters and identify abnormal conditions in real time. The system should process sensor data, compare the readings with predefined safety limits, and immediately generate visual or audible alerts whenever a dangerous condition is detected.

The proposed system aims to provide an additional layer of safety by reducing dependence on manual observation and helping workers and supervisors respond quickly to hazardous situations. The system can also be enhanced with IoT connectivity to transmit safety information remotely, maintain historical records, and support better safety management at construction sites.

PROBLEM SOLUTION

The proposed Worker Health and Safety Monitoring System provides an automated solution for continuously monitoring important safety parameters at construction sites. Sensors are used to collect real-time information about the worker and surrounding environmental conditions, while a microcontroller processes the sensor data and compares it with predefined safety limits.

When all monitored parameters remain within the safe range, the system indicates normal working conditions. If any parameter exceeds the predefined limit, the system automatically activates a buzzer and LED indicator to alert the worker and supervisor. This enables faster response to unsafe situations and helps prevent accidents and health-related risks.

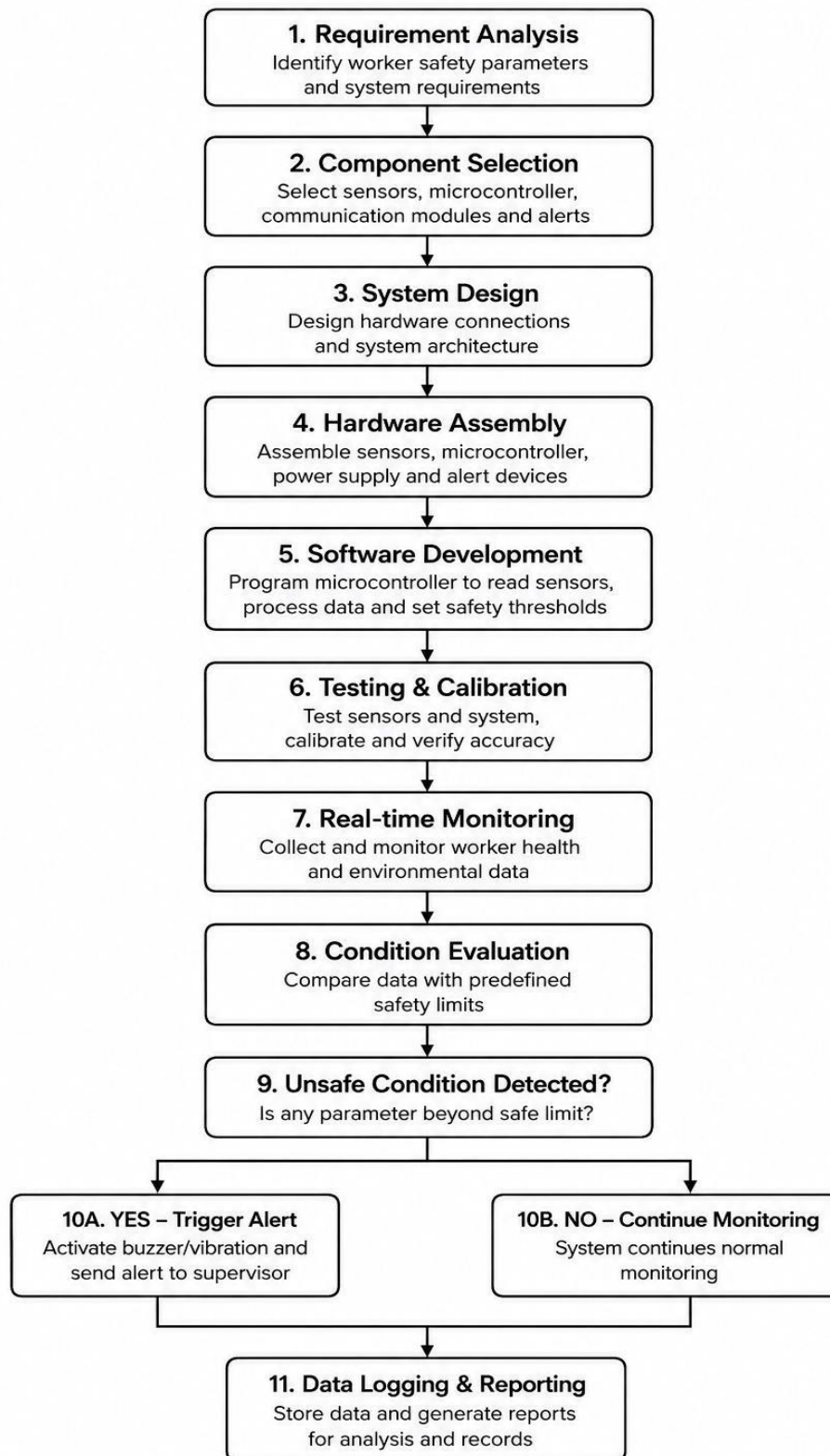
- Continuously monitors worker safety parameters.
- Detects unsafe environmental conditions in real time.
- Uses sensors for automatic safety monitoring.
- Processes sensor data using a microcontroller.
- Compares readings with predefined safety limits.
- Provides immediate visual and audible alerts.
- Reduces dependence on continuous manual supervision.
- Helps workers identify dangerous conditions quickly.
- Supports faster emergency response.
- Can store safety data for future analysis.
- Can be enhanced with IoT for remote monitoring and notifications.
- Provides a low-cost and practical safety solution for construction industries.

SYSTEM ARCHITECTURE

The Worker Health and Safety Monitoring System consists of sensors, a microcontroller, an alert unit, and a monitoring unit. The sensors continuously collect important worker and environmental safety parameters such as temperature, humidity, motion, and emergency conditions. The collected data is sent to the microcontroller, which processes the readings and compares them with predefined safety limits. When the monitored parameters are within the safe range, the system indicates normal working conditions. If any parameter exceeds the specified limit, the microcontroller activates the LED and buzzer to provide an immediate warning to the worker and supervisor. The system can also use a display to show real-time safety information and can be connected to an IoT platform for remote monitoring, data storage, and emergency notifications. This architecture enables continuous monitoring and faster response to unsafe conditions at construction sites.

The system architecture is designed to provide continuous and automatic safety supervision at the construction site. Sensor readings are collected at regular intervals and processed by the controller to identify changes in working conditions. Based on the detected values, the system can classify the

situation as safe or unsafe and immediately provide an appropriate alert. The architecture can also be expanded by adding wireless communication, allowing supervisors to receive safety information remotely and maintain records of hazardous events. This makes the system suitable for improving workplace safety, reducing response time, and supporting effective safety management in construction environments.

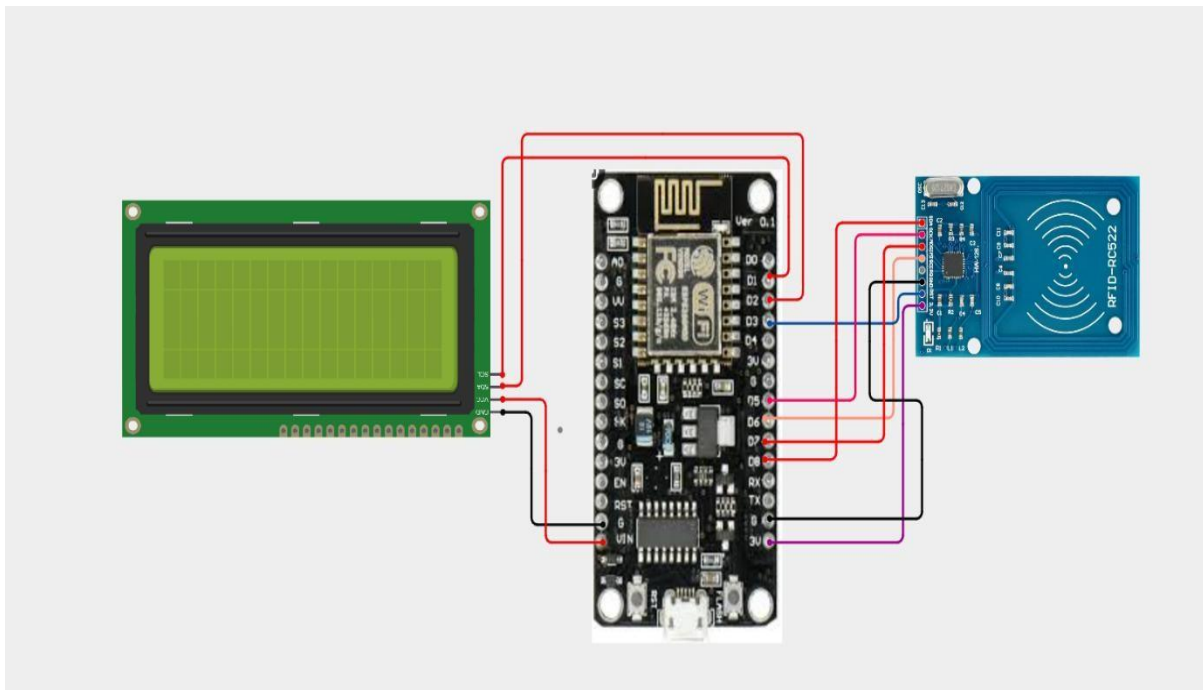


CIRCUIT TABLE

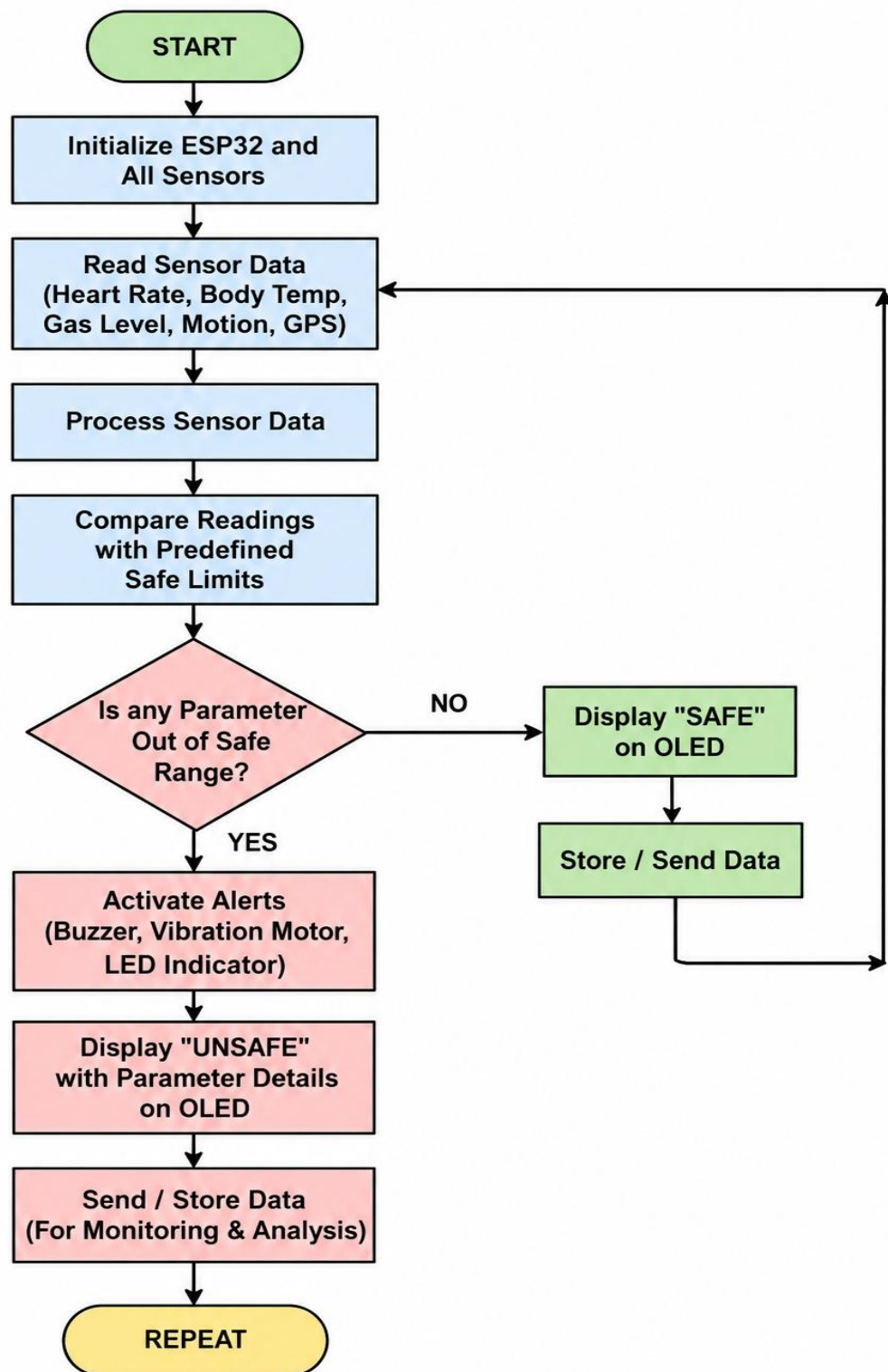
Component	ESP-8266 PIN
Heart Rate Sensor	GPIO 34
DS18B20	GPIO 4
MQ-135	GPIO 35
MPU6050	GPIO 21, 22
GPS	GPIO 16, 17
Buzzer	GPIO 25
Vibration Moto	GPIO 26
LED	GPIO 27
OLED	GPIO 21, 22
POWER	VIN , GND

- **ESP8266** → Processes the sensor and RFID data.
- **I2C LCD** → Displays material name and available quantity.
- **Buzzer** → Provides a low-stock or warning alert.
- **LED** → Indicates system/status conditions.

CIRCUIT DESIGN



ALGORITHM



- Start the system and initialize the ESP32 and all connected sensors.
- Collect data from the heart-rate, body-temperature, gas, accelerometer, and environmental sensors.
- Process the sensor data using the ESP32.
- Compare the readings with predefined safety limits.
- Check for unsafe conditions such as abnormal heart rate, high temperature, hazardous gas, or worker fall.
- If unsafe conditions are detected, activate the buzzer and vibration motor.
- Send an emergency alert and worker location to the supervisor through wireless communication.
- If conditions are normal, continue monitoring without triggering an alert.
- Display and store the sensor data for monitoring and future analysis.
- Repeat the monitoring process continuously.

CODING

```
#include <Wire.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <TinyGPSPlus.h>

// ----- PIN DEFINITIONS -----

#define HEART_PIN    34
#define TEMP_PIN     4
#define GAS_PIN      35

#define BUZZER_PIN   25
#define VIBRATION_PIN 26
#define LED_PIN      27
```



```
#define GPS_RX      16
#define GPS_TX      17

// ----- OLED -----
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// ----- SENSORS -----
OneWire oneWire(TEMP_PIN);
DallasTemperature tempSensor(&oneWire);

Adafruit_MPU6050 mpu;

TinyGPSPlus gps;
HardwareSerial GPSserial(2);

// ----- SAFETY LIMITS -----
float MAX_TEMPERATURE = 38.0;
int MAX_GAS = 2500;

// MPU6050 movement/fall threshold
float ACCEL_LIMIT = 3.0;

// ----- VARIABLES -----
float bodyTemperature;
int heartValue;
int gasValue;

float ax, ay, az;
float acceleration;
```

```
bool unsafeCondition = false;

void setup() {

    Serial.begin(115200);

    // Output pins
    pinMode(BUZZER_PIN, OUTPUT);
    pinMode(VIBRATION_PIN, OUTPUT);
    pinMode(LED_PIN, OUTPUT);

    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(VIBRATION_PIN, LOW);
    digitalWrite(LED_PIN, LOW);

    // Temperature sensor
    tempSensor.begin();

    // I2C
    Wire.begin(21, 22);

    // OLED
    if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
        Serial.println("OLED not found!");
    }

    display.clearDisplay();
    display.setTextColor(SSD1306_WHITE);
    display.setTextSize(1);
    display.setCursor(0, 0);
```

```
display.println("WORKER SAFETY");
display.println("SYSTEM STARTING...");
display.display();

// MPU6050
if (!mpu.begin()) {
    Serial.println("MPU6050 not detected!");
} else {
    Serial.println("MPU6050 Ready");
}

// GPS
GPSserial.begin(9600, SERIAL_8N1, GPS_RX, GPS_TX);

delay(2000);

Serial.println("Worker Safety System Ready");
}

// -----
// READ GPS
// -----

void readGPS() {

    while (GPSserial.available()) {
        gps.encode(GPSserial.read());
    }

    if (gps.location.isValid()) {
```

```
Serial.print("Latitude: ");
Serial.println(gps.location.lat(), 6);

Serial.print("Longitude: ");
Serial.println(gps.location.lng(), 6);
}
}

// -----
// READ ALL SENSORS
// -----

void readSensors() {

    // Heart-rate sensor
    heartValue = analogRead(HEART_PIN);

    // Temperature
    tempSensor.requestTemperatures();
    bodyTemperature = tempSensor.getTempCByIndex(0);

    // Gas sensor
    gasValue = analogRead(GAS_PIN);

    // MPU6050
    sensors_event_t a, g, temp;
    mpu.getEvent(&a, &g, &temp);

    ax = a.acceleration.x;
```

```
ay = a.acceleration.y;
```

```
az = a.acceleration.z;
```

```
acceleration = sqrt(
```

```
    ax * ax +
```

```
    ay * ay +
```

```
    az * az
```

```
);
```

```
}
```

```
// -----
```

```
// CHECK SAFETY CONDITIONS
```

```
// -----
```

```
void checkSafety() {
```

```
    unsafeCondition = false;
```

```
    // High body temperature
```

```
    if (bodyTemperature > MAX_TEMPERATURE) {
```

```
        unsafeCondition = true;
```

```
        Serial.println("WARNING: HIGH TEMPERATURE");
```

```
    }
```

```
    // High gas concentration
```

```
    if (gasValue > MAX_GAS) {
```

```
        unsafeCondition = true;
```

```
        Serial.println("WARNING: HIGH GAS LEVEL");
```

```
    }
```

```
// Sudden movement / possible fall
if (acceleration > ACCEL_LIMIT) {
    unsafeCondition = true;
    Serial.println("WARNING: SUDDEN MOVEMENT");
}

// Activate alerts
if (unsafeCondition) {

    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
    digitalWrite(VIBRATION_PIN, HIGH);

} else {

    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(VIBRATION_PIN, LOW);
}
}

// -----
// OLED DISPLAY
// -----

void showDisplay() {

    display.clearDisplay();

    display.setTextSize(1);
```

```
display.setCursor(0, 0);
```

```
display.println("WORKER SAFETY");
```

```
display.print("Temp: ");
```

```
display.print(bodyTemperature);
```

```
display.println(" C");
```

```
display.print("Heart: ");
```

```
display.println(heartValue);
```

```
display.print("Gas: ");
```

```
display.println(gasValue);
```

```
display.print("Accel: ");
```

```
display.println(acceleration, 1);
```

```
display.println();
```

```
if (unsafeCondition) {
```

```
    display.setTextSize(2);
```

```
    display.println("UNSAFE");
```

```
} else {
```

```
    display.setTextSize(2);
```

```
    display.println("SAFE");
```

```
}
```

```
display.display();
```

```
}
```

```
// -----
```

```
// SERIAL MONITOR
```

```
// -----
```

```
void printData() {
```

```
    Serial.println("-----");
```

```
    Serial.print("Heart Sensor : ");
```

```
    Serial.println(heartValue);
```

```
    Serial.print("Temperature : ");
```

```
    Serial.print(bodyTemperature);
```

```
    Serial.println(" C");
```

```
    Serial.print("Gas Level  : ");
```

```
    Serial.println(gasValue);
```

```
    Serial.print("Acceleration : ");
```

```
    Serial.println(acceleration);
```

```
    if (unsafeCondition)
```

```
        Serial.println("STATUS    : UNSAFE");
```

```
    else
```

```
        Serial.println("STATUS    : SAFE");
```

```
    Serial.println("-----");
```

```
}
```



```
// -----  
// MAIN LOOP  
// -----  
  
void loop() {  
  
    // Read sensor values  
    readSensors();  
  
    // Read GPS data  
    readGPS();  
  
    // Check safety conditions  
    checkSafety();  
  
    // Display values  
    showDisplay();  
  
    // Serial monitor  
    printData();  
  
    // Wait before next reading  
    delay(1000);  
}
```

SYSTEM WORKING

The Worker Health and Safety Monitoring System continuously monitors the worker's health and surrounding environmental conditions using multiple sensors connected to the ESP32. The heart rate sensor measures the worker's heart activity, while the DS18B20 measures body temperature.

The MQ-135 detects harmful gas levels, and the MPU6050 monitors sudden movement or possible fall conditions. The GPS module provides the worker's location during an emergency.

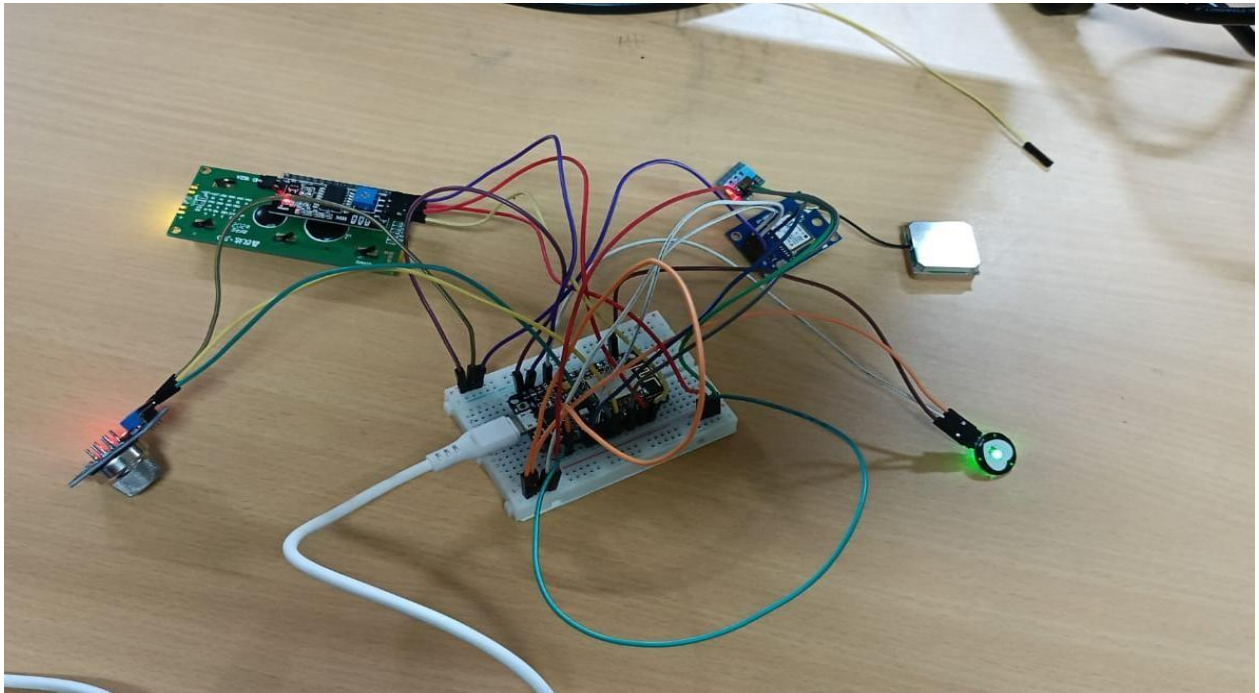
The ESP32 receives and processes all sensor readings and compares them with predefined safety limits. When the readings are within the safe range, the system displays “**SAFE**” on the OLED display and continues normal monitoring. If any parameter exceeds its safe limit, the ESP32 identifies the condition as unsafe and activates the **buzzer, vibration motor, and LED indicator** to immediately alert the worker.

The OLED display provides real-time information about the monitored parameters and safety status. The GPS module can provide location information during an emergency, allowing the worker's position to be identified. The system continuously repeats this monitoring process, providing an automated safety mechanism that helps workers and supervisors respond quickly to hazardous conditions at construction sites.

BILL OF MATERIAL

COMPONENT	QUANTITY
ESP8266 NodeMCU	1
Pulse Sensor	1
DHT11 Sensor	1
MQ2 Gas Sensor	1
Buzzer	1
OLED Display	1
GPS Module (NEO-6M	1
Breadboard	1
Jumper Wires	1 SET

PROTOTYPE IMPLEMENTATION



- ESP32 Setup – The ESP32 is used as the main controller for processing all sensor data.
- Sensor Integration – Connect the MAX30102, DS18B20, MQ-135, MPU6050, DHT22, and GPS sensors to the ESP32.
- Health Monitoring – The MAX30102 measures heart rate and the DS18B20 monitors body temperature.
- Environmental Monitoring – The MQ-135 detects harmful gases, while the DHT22 measures temperature and humidity.
- Fall Detection – The MPU6050 detects sudden movements, impacts, and possible worker falls.
- Location Tracking – The NEO-6M GPS module provides the worker's location during emergencies.
- Display Unit – The OLED display shows sensor readings and the current safety status.
- Alert System – A buzzer and vibration motor warn the worker when an unsafe condition is detected.
- Wireless Communication – ESP32 transmits safety information to a supervisor or monitoring system.
- Power Supply – A rechargeable battery provides power to the wearable prototype.

- Programming – The ESP32 is programmed to continuously read sensor values and compare them with predefined safety limits.
- Testing – The prototype is tested under simulated conditions such as high temperature, gas detection, abnormal movement, and fall detection.
- Final Prototype – All components are assembled into a compact wearable unit suitable for mounting on a safety helmet, vest, or belt.

RESULT AND PERFORMANCE ANALYSIS

The Worker Health and Safety Monitoring System was successfully designed and tested to monitor important worker and environmental safety parameters in real time. The ESP32 successfully collected data from the heart rate sensor, body temperature sensor, MQ-135 gas sensor, MPU6050 accelerometer, and GPS module. The sensor readings were processed and compared with predefined safety limits to determine the worker's safety condition.

During normal operating conditions, the system displayed the monitored parameters and indicated “SAFE” on the OLED display. When an unsafe condition was simulated by increasing the temperature, gas level, or detecting sudden movement, the system responded by activating the **buzzer, vibration motor, and LED indicator**. This provided immediate visual, audible, and physical warnings to the worker.

The system demonstrated continuous monitoring with automatic alert generation and showed good response during testing. The OLED display provided real-time information, while the GPS module supported location monitoring during emergency conditions. Overall, the prototype demonstrates that the proposed system can improve worker safety by providing early detection of hazardous conditions and reducing dependence on continuous manual supervision. The system can be further improved by adding cloud monitoring, mobile notifications, accurate BPM calculation, and emergency alerts to supervisors.

